

When Two Sigma, a quantitative hedge fund managing over \$60 billion, built their Beaker notebook (now an open-source project), they chose Python as one of the primary languages precisely because of its integration capabilities.

Here's a simplified example of how Python can connect to a trading API:

```
1. # Simple example of connecting to a trading API
2. import requests
3.
4. # Get the current price of Apple stock
5. response = requests.get("<https://api.example.com/stocks/AAPL/price>")
6. apple_price = response.json()["price"]
7.
8. print("Current Apple stock price:", apple_price)
9.
```

Real-World Algorithmic Trading Strategies

Understanding the specific types of trading strategies that utilize Python helps illustrate why the language has become so indispensable. Let's explore some of the major categories of algorithmic trading strategies and how Python facilitates their implementation.

High-Frequency Trading (HFT)

High-frequency trading involves executing a large number of orders at extremely high speeds, often holding positions for milliseconds or microseconds. While the core execution engines for HFT are typically written in lower-level languages like C++ for maximum speed, Python often plays crucial roles in:

- Strategy research and backtesting
- Data analysis and pattern identification
- Risk management and position monitoring
- Post-trade analysis